

Superposition and Quantum Gates

Week 2: Quantum Computing Fundamentals

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Today's Journey

- 1 **Mathematical Superposition** - Complex amplitudes and the Born rule
- 2 **Bloch Sphere Visualization** - Geometric understanding of quantum states
- 3 **Essential Quantum Gates** - The building blocks of quantum computation
- 4 **Circuit Construction** - Reading and designing quantum algorithms
- 5 **Multi-qubit Systems** - Scaling up to practical quantum computing

Review: From Classical to Quantum

Last Week's Key Insights:

- Classical coins: definitively heads OR tails
- Quantum coins: superposition - both heads AND tails simultaneously
- Exponential advantage: n quantum coins represent all 2^n possibilities

Today's Goal: Mathematical precision and practical tools for quantum computation

General Qubit State:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Probability Amplitudes: α and β are complex numbers

$$\alpha = a + bi, \quad \beta = c + di$$

The Born Rule (Measurement):

- Outcome $|0\rangle$ with probability $|\alpha|^2 = a^2 + b^2$
- Outcome $|1\rangle$ with probability $|\beta|^2 = c^2 + d^2$

Key Insight: Probabilities come from squared magnitudes!

Normalization Condition

Physical Constraint:

$$|\alpha|^2 + |\beta|^2 = 1$$

Examples:

- ✓ Valid: $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
 - Check: $|\frac{1}{\sqrt{2}}|^2 + |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} + \frac{1}{2} = 1$
- Invalid: $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle$
 - Check: $|\frac{1}{2}|^2 + |\frac{1}{2}|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \neq 1$

The Magic of Quantum Interference

Two states with identical measurement probabilities:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{-1}{\sqrt{2}}|1\rangle$$

But different interference behavior:

$$|\psi_1\rangle + |\psi_2\rangle = \sqrt{2}|0\rangle + 0|1\rangle$$

- $|1\rangle$ components canceled out (destructive interference)
- $|0\rangle$ components added up (constructive interference)

This is how quantum algorithms work!

Two Qubits:

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$

Normalization: $|\alpha_{00}|^2 + |\alpha_{01}|^2 + |\alpha_{10}|^2 + |\alpha_{11}|^2 = 1$

General Pattern:

- n qubits require 2^n complex amplitudes
- This is why quantum systems are powerful and hard to simulate!

Mathematical Mapping:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

Spherical Coordinates:

- θ : polar angle (0 to π) - determines measurement probabilities
- ϕ : azimuthal angle (0 to 2π) - determines phase relationships

Key Landmark States:

- North Pole (0, 0): $|0\rangle$
- South Pole (π , 0): $|1\rangle$
- Equator ($\pi/2$, 0): $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- Equator ($\pi/2$, π): $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

Measurement Probabilities:

- Probability of $|0\rangle = \cos^2(\theta/2)$
- Depends only on "height" on the sphere

Phase Relationships:

- Same latitude = same measurement probabilities
- Different longitude = different phases
- Phase differences matter for interference!

Quantum Gates:

- Every single-qubit gate = rotation of Bloch sphere
- Powerful geometric intuition!

Limitation: Only works for single qubits

The Pauli Gates

X Gate (Quantum NOT):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle$$

Bloch sphere: 180° rotation around X-axis

Y Gate:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle$$

Bloch sphere: 180° rotation around Y-axis

Z Gate (Phase Flip):

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle$$

Bloch sphere: 180° rotation around Z-axis

The Hadamard Gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Magic Actions:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$$

Remarkable Property: H is its own inverse! $H \cdot H = I$

- $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$

The "superposition creator"

S Gate (Phase gate):

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad S|0\rangle = |0\rangle, \quad S|1\rangle = i|1\rangle$$

Bloch sphere: 90° rotation around Z-axis

T Gate ($\pi/8$ gate):

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \quad T|0\rangle = |0\rangle, \quad T|1\rangle = e^{i\pi/4}|1\rangle$$

Bloch sphere: 45° rotation around Z-axis

Purpose: Precise control over interference patterns

Gate Composition

Sequential Application: Apply gate A then gate $B = B \cdot A$

Worked Example: $H \rightarrow Z \rightarrow H$ sequence

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (1)$$

$$\xrightarrow{Z} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (2)$$

$$\xrightarrow{H} |1\rangle \quad (3)$$

Matrix Calculation: $H \cdot Z \cdot H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$

Amazing Result: $HZH = X!$

Circuit Structure:

- Time flows left to right
- Each horizontal line = one qubit
- Gates = boxes or symbols on lines
- Qubits start in $|0\rangle$ unless specified

Standard Symbols:

- X gate: $\oplus$ (circle with plus)
- H gate: [H] (box with H)
- Z gate: [Z] (box with Z)
- Measurement: meter symbol

Circuit Analysis Method

Step-by-Step Process:

- 1 **Initialize:** Start with $|00\dots 0\rangle$
- 2 **Apply gates:** Work left to right
- 3 **Calculate:** Track quantum state evolution
- 4 **Measure:** Compute final probabilities

Example: $|0\rangle \xrightarrow{B} H \xrightarrow{B} Z \xrightarrow{B} H \xrightarrow{B} ?$

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad (4)$$

$$\xrightarrow{Z} \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \quad (5)$$

$$\xrightarrow{H} |1\rangle \quad (6)$$

This circuit converts $|0\rangle \rightarrow |1\rangle$ deterministically!

Key Rules:

- **No Cloning:** Cannot copy arbitrary quantum states
- **Reversibility:** All quantum gates (except measurement) are reversible
- **Unitarity:** Gate matrices preserve normalization
- **Measurement:** Destroys superposition, gives classical info

Design Strategy: Engineer interference to amplify correct answers and suppress wrong ones

Product States vs Entangled States

Product States (Separable): $|\psi\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$
 $= ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$

Example: $|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

Entangled States: CANNOT be written as products

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Next Week: Entangled states exhibit quantum correlations with no classical analog!

Single-Qubit Gates on Multi-Qubit Systems

Tensor Product Formalism: Apply X to first qubit of two-qubit system:

$$X \otimes I = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

Practical Effect: Gate acts on target qubit, leaves others unchanged

Equal Superposition: Apply H to both qubits

$$|00\rangle \xrightarrow{H \otimes H} \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

The Recipe for Quantum Advantage

Essential Ingredients:

- ➊ **Superposition Creation** → H gates explore all possibilities
- ➋ **Amplitude Manipulation** → Various gates modify amplitudes
- ➌ **Interference Engineering** → Wrong answers cancel, right answers amplify
- ➍ **Measurement** → Extract classical information

Next Week: Entanglement adds quantum correlations for even more sophisticated algorithms

Preview: Deutsch's Algorithm (Week 4)

The Problem: Is a function constant or balanced?

Classical: Must check function twice (worst case)

Quantum Strategy:

- 1 **Superposition:** Query all inputs simultaneously with $|+\rangle$
- 2 **Oracle:** Apply unknown function
- 3 **Interference:** H gate makes constant/balanced interfere differently
- 4 **Measurement:** One measurement gives the answer!

Result: Quantum advantage - one query instead of two!

What We've Accomplished Today

Mathematical Foundation:

- ✓ Superposition and probability amplitudes
- ✓ Complex numbers and quantum interference
- ✓ Normalization and Born rule

Geometric Intuition:

- ✓ Bloch sphere visualization
- ✓ Quantum gates as rotations

Practical Skills:

- ✓ Fundamental quantum gates
- ✓ Circuit analysis and design
- ✓ Multi-qubit systems

Algorithmic Thinking:

- ✓ Engineering quantum interference
- ✓ Foundation for quantum algorithms

Next Week: Entanglement

Einstein's "Spooky Action at a Distance"

We'll Learn:

- How to create quantum correlations between qubits
- The CNOT gate and two-qubit operations
- Bell states and their remarkable properties
- Why entanglement is the defining feature of quantum mechanics

The Complete Foundation: After next week, you'll have all tools needed to implement quantum algorithms!

You're now thinking precisely about quantum computation!

Thank you!

Questions?

Next week: Entanglement and quantum correlations